

Implement Client Script & UI Policy (Incident)

ServiceNow | Incident Management

Team Details

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Topic: Implement Client Script & UI Policy (Incident)

Project Overview

Problem Statement

Incident records require consistent and accurate data entry to support effective triage, routing and resolution. Relying only on user awareness and manual checks can lead to incomplete, inconsistent or incorrect data being submitted, which affects reporting accuracy, SLA compliance and overall service quality. To address this, conditional field behaviour and validation must be enforced directly at the user interface level. Standardized UI Policies and Client Scripts control field visibility, mandatory status and save-time validation, so that Incident records meet the defined business rules before they are submitted.

Objective

The objective of this project is to demonstrate how ServiceNow client-side controls can enforce data integrity on Incident records. The implementation shows how UI Policies and Client Scripts can dynamically make fields mandatory, auto-populate values, control field behaviour and prevent record submission when required conditions are not met. The solution ensures that Incidents are created with complete and valid information, improving data quality and consistency across the platform.

Skills

- Incident Management
- UI Policy
- UI Policy Actions
- Client Scripts
- Form Validation

Milestones

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Milestone 1: Create UI Policy on Incident

Activity 1: Create UI Policy

1. Log in to the ServiceNow instance with administrative or configuration access.
2. Navigate to **System UI > UI Policies**.
3. Click **New** to create a new UI Policy record.
4. Fill in the details: **Short description**: High Impact Control (the policy applies additional controls when an Incident has a high impact); **Table**: Incident; **Active**: true.
5. In the Conditions section set **Field**: Impact, **Operator**: is, **Value**: 1 - High, so that the policy triggers only when the Incident impact is High.
6. Set **Reverse if false** to true, so that field changes applied by the policy are reverted when the condition is not met.
7. Create a UI Policy Action on the field **Assignment group** and set **Mandatory** = **True** and **Read only** = **False**.
8. Review the configuration and click **Submit** to save the UI Policy.

Field	Value
Short description	High Impact Control
Table	Incident [incident]
Active	True
Condition	Impact is 1 - High
Reverse if false	True
Action field	Assignment group (Mandatory = True, Read only = False)

Output

Short description	Table	Conditions	Reverse if false	On load	Updated	Order
UI Policy to hide related feature flags when hidePagination set to true	UX List [sys_ux_list]	hide_pagination=true^EQ	true	true	2021-08-16 14:02:50	
Mark fields as mandatory without DB override	Security tasks [sn_vsc_security_task]		false	true	2025-01-28 14:28:42	
Always Hide Read-only Related List Field	Relationship [sys_relationship]	sys_idISNOTEMPTY^EQ	false	true	2015-08-21 17:28:31	
Hide meta fields	label [sc_ic_question]		true	true	2014-01-21 12:43:35	
Set Table column ReadOnly	To-dos Widget Mapping [sn_hr_sp_todos_widget_mapping]		true	true	2018-08-02 15:16:10	
Clear save signature before action execution value - when precondition is not Signature	Function [sys_sg_button]	preconditions!=signature^EQ	true	true	2019-04-18 07:01:21	
Content vs parameters	HTTP Method [sys_rest_message_fn]	http_method=get^ORhttp_method=delete^ORh...	true	true	2016-12-14 16:50:13	

servicenow

All

UI pol

FAVORITES

No Results

ALL RESULTS

System Mobile

Mobile UI

Mobile UI Policies

Service Catalog

Catalog Administration

Catalog UI Policies

System Diagnostics

Session Debug

Debug UI Policies

Disable UI Policies Debug

System UI

UI Policies

MID Server

MID Security Policy

System Web Services

SOAP Security Policies

Favorites History Workspaces Admin

UI Policy - High Impact Control

Search

UI Policy

High Impact Control

View: Advanced

Update Delete

The field specified in the UI policy action is present on the specified form

More info

Table Incident [incident]

Application Global

Active

* Short description High Impact Control

Order 100

When to Apply

Script

Conditions

Add Filter Condition

Add OR Clause

Impact is 1 - High

AND OR

If selected, the UI Policy applies to all form views; otherwise the UI Policy is view-specific

Apply the UI policy actions when the form is loaded and when the user changes values on the form

Global

Reverse the effects of the UI policy actions when the Conditions evaluate to false

Reverse if false

On load

Tables that extend the specified Table inherit this UI Policy

Inherit

servicenow

All

UI pol

FAVORITES

No Results

ALL RESULTS

System Mobile

Mobile UI

Mobile UI Policies

Service Catalog

Catalog Administration

Catalog UI Policies

System Diagnostics

Session Debug

Debug UI Policies

Disable UI Policies Debug

System UI

UI Policies

MID Server

MID Security Policy

System Web Services

SOAP Security Policies

Favorites History Workspaces Admin

UI Policy Action - assignment_group

Search

UI Policy Action

assignment_group

Update Delete

UI policy actions specify exactly what actions to take on a specified field. The conditions specified in the UI policy determine when these actions are triggered. More Info

UI policy High Impact Control

Table Incident [incident]

* Field name Assignment group

Application Global

Mandatory True

Visible Leave alone

Read only False

Clear the field value

Update Delete

Milestone 2: Create UI Policy Action - Urgency

Activity 1: Configure the Urgency Field

1. Open the **High Impact Control** UI Policy created earlier and scroll to the **UI Policy Actions** related list.
2. Click **New** to create a UI Policy Action.
3. Set **Field name**: Urgency. This is the field to be controlled.
4. Set **Read only**: true, so the Urgency field becomes read-only whenever the condition (Impact = High) is met. Users can see the field but cannot change its value.
5. Leave **Visible** as is (do not change), so the field remains visible on the form.
6. Click **Submit**. The action is now linked to the High Impact Control UI Policy and applies automatically when an Incident has High Impact.

Optional Verification

- Open an Incident record and set Impact to High.
- Check that the Urgency field becomes read-only.
- Change Impact to a different value to confirm the field reverts.

Output

The screenshot shows the ServiceNow interface for configuring a UI Policy Action. The left sidebar displays a navigation tree with categories like System Mobile, Service Catalog, System Diagnostics, System UI, MID Server, and SOAP Security Policies. The main content area is titled 'UI Policy Action - urgency' and includes a search bar and action buttons (Update, Delete). Below the title, a blue banner states: 'UI policy actions specify exactly what actions to take on a specified field. The conditions specified in the UI policy determine when these actions are triggered. [More Info](#)'. The configuration fields are as follows:

Field	Value
UI policy	High Impact Control
Table	Incident [incident]
* Field name	Urgency
Application	Global
Mandatory	Leave alone
Visible	Leave alone
Read only	True
Clear the field value	☐

At the bottom of the configuration area, there are 'Update' and 'Delete' buttons.

Milestone 3: Create onChange Client Script

Activity 1: Create Client Script

1. Go to **System UI > Client Scripts** and click **New**.
2. **Name:** Auto set urgency for high impact; **Table:** Incident; **Type:** onChange; **Field name:** Impact; **Active:** true.
3. Paste the script given below in the script editor and click **Submit**.
4. Verify: open an Incident and change Impact to High. The Urgency field should update automatically.

Script

```
function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue == '') {
        return;
    }
    if (newValue == '1') { // High Impact
        g_form.setValue('urgency', '1');
        g_form.showFieldMsg('impact',
            'High impact incidents must have High urgency and an assignment group.',
            'info');
    }
}
```

Output

Name	Active	Table	Application	View	Type	Updated
Maximum Duration Range Check	true	Transaction Quota Rule [sysrule_quota]	Global		onChange	2012-05-30 14:47:01
Create default value	true	Response Template Channel [sys_template_intp_channel]	Templated Snippets		onChange	2020-09-22 17:06:27
Show overlay field based on data labels	true	Widgets [sys_widgets]	Global		onChange	2021-04-23 02:59:51
Set protocol based on auth type	false	HTTP(s) Connection [http_connection]	Global		onChange	2017-06-02 15:49:24
Set default values for ODC PACE flow	true	OAuth ODC Entry [oauth_ode_entry]	Global		onChange	2020-02-28 00:59:58
Make notification read only	true	Additional Notification Recipients [sys_notification_recipient]	Global		onload	2021-05-13 04:51:51
Deprecated - SN - Make serial number man	false	Asset [sys_asset]	Global		onChange	2025-03-06 00:42:54
Validate Stateful Name	true	MDK K8s Deployment Request [mdk_k8s_deployment]	Global		onChange	2022-10-09 23:38:07
Set Use current record if delete	true	Action Item [sys_action_item]	Global		onChange	2019-04-14 07:30:02
Warning when no destination type enabled	true	Notification [sys_notification]	Global		onload	2021-05-13 04:51:51
Sync Status Change Handler	true	Advanced Portal Navigation [sys_portal_accessible_navigation]	Employee Center		onChange	2024-12-23 03:29:15
Populate, hide/show relation core table	true	Finding Constraint [sys_finding_def_constraint]	Global		onChange	2022-09-12 07:27:29
Mark type read only	true	System Address Filters [sys_email_system_filter]	Global		onload	2020-02-17 02:09:09
Populate Process Definition	true	Activity [sys_act_activity]	Global		onload	2022-08-22 13:59:47
Populate downgrade from license	true	Software Upgrade and Downgrades [smdb_mdm_downgrade_model]	Global		onChange	2014-02-03 16:40:33
Set Activity Definition Variables	true	Workflow UI Policy Action [wf_ui_policy_action]	Global		onChange	2016-02-05 07:22:26
Set file name extn	true	Export Set [sys_export_set]	Global		onload	2017-05-23 16:49:28

Client Script - Auto set urgency for high impact

Name: Auto set urgency for high impact

Table: Incident

UI Type: Globalize

Type: onChange

Field name: Impact

Application: Global

Active: ☒

Inherited: ☐

Global: ☒

Description:

Messages:

Script:

```
1 function onChange(control, oldValue, newValue, isLoading) {
2   if (isLoading || newValue == '') {
3     return;
4   }
5   if (newValue == '1') { // High Impact
6     g_form.setValue('urgency', '1');
7     g_form.showFieldMsg(
8       'impact',
9       'High impact incidents must have High urgency and an assignment group.',
10      'info'
11    );
12  }
13 }
14
15
```

Isolate script: ☒

Update Delete

Milestone 4: Create onSubmit Client Script (Save Validation)

Activity 1: Create Client Script

1. Go to **System UI > Client Scripts** and click **New**.
2. **Name:** Prevent save if Assigned To missing; **Table:** Incident; **Type:** onSubmit; **Active:** true.
3. Paste the script given below. It checks whether Assigned To is empty for a High impact Incident and prevents the save if it is.
4. Click **Submit**.
5. Verify: try saving an Incident (Impact = High) without Assigned To. The save is prevented and an error is displayed.

Script

```
function onSubmit() {  
    if (g_form.getValue('impact') == '1' &&  
        g_form.getValue('assigned_to') == '') {  
        g_form.showErrorBox(  
            'assigned_to',  
            'Assigned To is mandatory for High impact incidents.'  
        );  
        return false;  
    }  
    return true;  
}
```

Output

The screenshot shows the ServiceNow interface for configuring a Client Script. The left sidebar contains the navigation menu with 'Client Scripts' selected. The main area displays the configuration for a Client Script named 'Prevent save if assigned to missing'. The configuration includes the following fields:

- Name:** Prevent save if assigned to missing
- Table:** Incident [Incident]
- UI Type:** Desktop
- Type:** onSubmit
- Application:** Global
- Active:** ☒
- Inherited:** ☐
- Global:** ☒

The Description and Messages fields are empty. The Script field contains the following code:

```
1 function onSubmit() {  
2     if (g_form.getValue('impact') == '1' &&  
3         g_form.getValue('assigned_to') == '') {  
4  
5         g_form.showErrorBox(  
6             'assigned_to',  
7             'Assigned To is mandatory for High Impact Incidents.'  
8         );  
9         return false;  
10    }  
11    return true;  
12 }  
13  
14
```

The 'Isolate script' checkbox is checked. At the bottom, there are 'Update' and 'Delete' buttons.

Milestone 5: Create onCellEdit Client Script

Activity 1: Create Client Script

1. Go to **System UI > Client Scripts** and click **New**.
2. **Name:** Prevent state change via list edit; **Table:** Incident; **Type:** onCellEdit; **Field name:** State; **Active:** true.
3. Paste the script given below. It prevents users from changing the State field directly from the list view.
4. Click **Submit**.
5. Verify: open the Incident list and try to change State directly in the list. The change is blocked.

Script

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) {  
  
    alert('State cannot be changed from the list. Please open the Incident.');
```

```
    // Cancel the list edit  
    callback(false, oldValues);  
}
```

Output

The screenshot shows the ServiceNow interface for creating a new Client Script. The left sidebar displays the navigation menu with 'Client Scripts' selected. The main area shows the configuration form for a new Client Script named 'Prevent state change via list edit'. The form includes fields for Name, Table (Incident), UI Type (Desktop), Type (onCellEdit), Field name (State), Application (Global), Active (checked), Inherited (unchecked), and Global (checked). The Script field contains the following code:

```
1 function onCellEdit(sysIDs, table, oldValues, newValue, callback) {  
2  
3     alert('State cannot be changed from the list. Please open the Incident.');
```

```
4  
5     // Cancel the list edit  
6     callback(false, oldValues);  
7 }  
8  
9
```

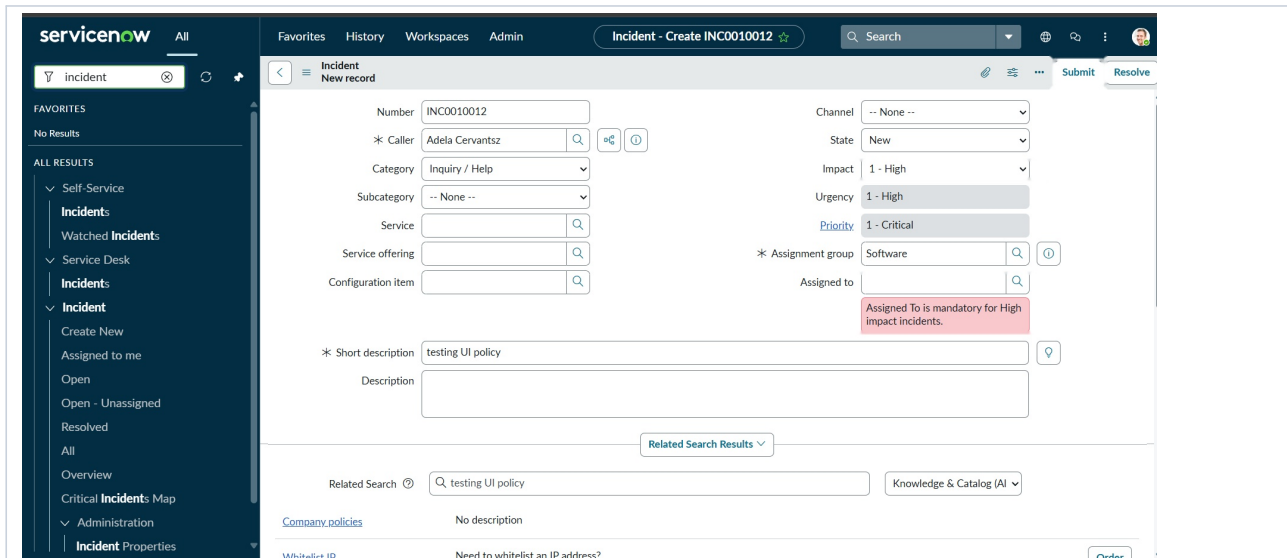
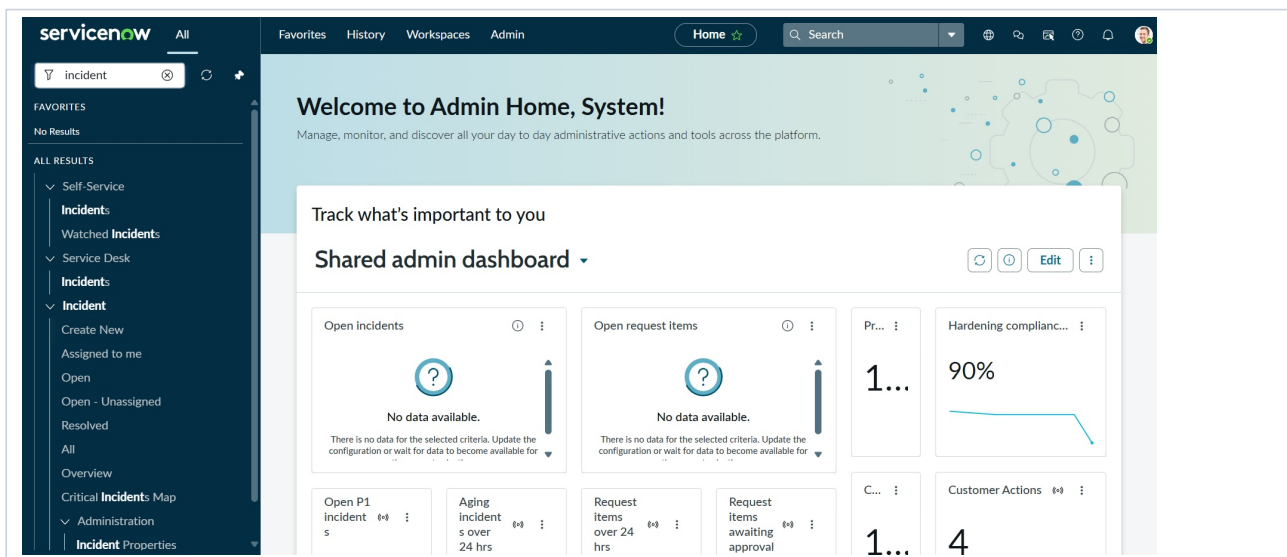
The 'Isolate script' checkbox is checked at the bottom.

Milestone 6: Testing the Configuration

Activity 1: Test Mandatory Enforcement

1. Go to **Incident > Create New** to open a new Incident form.
2. Select **High** for the Impact field. This triggers the UI Policy and Client Script behaviours.
3. Leave **Assigned To** empty to test the onSubmit Client Script.
4. Click **Submit** to attempt to save the record.
5. Verify: the Incident is not saved, and an error message states that Assigned To is mandatory for High impact incidents.

Output



Activity 2: Test Successful Save

1. Open the Incident form and select a user in the **Assigned To** field.
2. Click **Submit**.

3. Verify: the Incident saves successfully. Urgency is auto-set (onChange) and Urgency is read-only (UI Policy Action).

Output

Activity 3: Reverse Condition Test

1. Go to **Incident > Open** and select an Incident where Impact is High.
2. Change **Impact** from High to Medium to test Reverse if false and the Client Script behaviour.
3. Verify: Assigned To is no longer mandatory and Urgency is editable again (read-only action reversed).
4. Save the record. It saves without errors.

Output

Activity 4: Test List Edit Blocking

1. Go to **Incident > All** to view Incident records in a list.
2. Double-click the **State** field of any Incident to edit it directly in the list.

3. Verify: an alert states the State cannot be changed from the list, and the State value remains unchanged.

Output

The screenshot shows the ServiceNow Incidents list view. The table contains the following data:

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF:TEST1	System Administrator	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-22 14:05:48	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Opspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2016-08-10	admin

The screenshot shows the same ServiceNow Incidents list view as above, but with a modal dialog box overlaid. The dialog box contains the following text:

dev354432.service-now.com says
State cannot be changed from the list. Please open the Incident.

There is an 'OK' button at the bottom right of the dialog box.

Activity 5: Test Form-Based Update

1. Open the same Incident record on the Incident form.
2. Change the **State** field to a new value (for example In Progress).
3. Click **Update** to save.
4. Verify: the State change is saved successfully, and other fields such as Assigned To and Urgency still follow the configured rules.

Output

The image shows two screenshots of the ServiceNow interface. The top screenshot displays the 'Incident - INC0008111' form. The form includes fields for Number, Caller, Category, Subcategory, Service offering, Configuration item, Short description, Description, Country, and States. A context menu is open over the form, showing options like 'Save', 'Add to Visual Task Board', 'Copy Incident', 'Create Incident Task', 'Create Outage', 'Create Problem', 'Create Request', 'Create Child Incident', 'Create Normal Change', 'Create Standard Change', 'Create Emergency Change', 'Refresh Impacted Services', 'Metrics Timeline', 'Follow on Live Feed', 'Analyze Access', 'Configure', 'Export', 'View', 'Create Favorite', 'Copy URL', 'Copy sys_id', 'Show XML', 'History', and 'Reload form'. The bottom screenshot shows the 'Incidents' list view. The list table has columns for Number, Opened, Short description, Caller, Priority, State, Category, Assignment group, Assigned to, Updated, and Updated by. The incident INC0008111 is highlighted in blue.

Incident Form Details:

- Number: INC0008111
- Caller: System Administrator
- Category: Inquiry / Help
- Subcategory: -- None --
- Service offering:
- Configuration item:
- Short description: ATF - Test1
- Description:
- Country: -- None --
- States: -- None --

Incidents List View:

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2026-01-14 03:31:59	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	David Miller	3 -	New	Inquiry / Help	(empty)	(empty)	2016-08-10	admin

Conclusion

The Implement Client Script & UI Policy (Incident) micro project demonstrates how UI Policies and Client Scripts work together to enforce dynamic field behaviour, automate updates and prevent incorrect data submission on Incident forms. By using onChange, onSubmit and onCellEdit scripts alongside UI Policy actions, the project ensures that fields such as Assigned To and Urgency respond appropriately to changes in Impact, maintaining clean and consistent data.

This solution enforces mandatory fields and controls user interactions while remaining lightweight, efficient and easy to implement. It is well suited to short-duration micro projects and shows practical best practices for improving form usability and data integrity in ServiceNow.

Key Outcomes

- Assignment group becomes mandatory when Impact is High (UI Policy and Reverse if false).
- Urgency becomes read-only for High impact and is auto-set to High by the onChange script.
- Incidents with High impact cannot be saved without Assigned To (onSubmit).
- State cannot be changed through list editing (onCellEdit), only through the form.
- All behaviours were tested and produced the expected results.

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